

Test Case Generator using JIRA Integration

Implemented Vibe Coding to develop a Test Case generator with JIRA Integration.

Fetching user stories from JIRA via API:

The screenshot shows a Postman REST client interface. The request is a POST to the URL `https://suhasiniworkspace-40182683.atlassian.net/rest/api/3/search?jql`. The request body is a JSON object with the following structure:

```
1 {
2   "jql": "key = \"SCRUM-2\"",
3   "fields": [
4     "key",
5     "summary",
6     "description",
7     "status",
8     "reporter",
9     "details"
10  ]
11 }
```

The response is a 200 OK status with a response time of 1.79 s and a size of 2.27 KB. The response body is a JSON object with the following structure:

```
6 {
7   "self": "https://suhasiniworkspace-40182683.atlassian.net/rest/api/3/issue/10004",
8   "key": "SCRUM-2",
9   "fields": {
10     "summary": "Consultant Dashboard My Inpatients",
11     "description": {
12       "type": "doc",
13       "version": 1,
14       "content": [
15         {
16           "type": "paragraph",
17           "content": [
18             {
19               "type": "text",
20               "text": "As a user, I want to consultant dashboard my inpatients so that the related functionality works as expected."
21             }
22           ]
23         }
24       ]
25     }
26   }
27 }
```

User stories in JIRA for reference:

The screenshot shows the JIRA SCRUM board for the project "SCRUM". The board displays a list of user stories, each with a checkbox, a link to the story, a description, an assignee, a reporter, and a priority.

Work	Assignee	Reporter	Priority
☐ SCRUM-52 Employee Profile Employee Self	Unassigned	SG Suhasini Ganesh	None
☐ SCRUM-51 AHC Reports Reports AHC	Unassigned	SG Suhasini Ganesh	None
☐ SCRUM-50 Secretary Dashboard Doctor Calendar	Unassigned	SG Suhasini Ganesh	None
☐ SCRUM-49 Masters Others Approval Setup	Unassigned	SG Suhasini Ganesh	None
☐ SCRUM-48 Employee Profile Basic Details	Unassigned	SG Suhasini Ganesh	None
☐ SCRUM-47 Doctor Dashboard Op Dashboard	Unassigned	SG Suhasini Ganesh	None
☐ SCRUM-46 Reports Admin Reports Others	Unassigned	SG Suhasini Ganesh	None
☐ SCRUM-45 Consultant Dashboard My Appointments	Unassigned	SG Suhasini Ganesh	None

Prompt given:

Analyse my existing code base and Implement Jira integration

GO STEP BY STEP up on my approval proceed with next step

--[Mandatory] Backend first then proceed with Frontend

Instruction

--[MANDATORY] Implement Jira connectivity email, Api key should be configurable

--[CRITICAL] Fetch the Projects from Jira (Projects should be auto populated as dropdown)

--[CRITICAL] Fetch User stories for the project (user stories should be auto populated as dropdown)

REFER MY POSTMAN COLLECTION attached

Followed by chain of prompts:

Proceed with backend

Proceed with frontend (given after enabling backend by running npm run dev)

JIRA Configuration invalid

Fix the .env file

Final O/P:

The screenshot displays a VS Code editor with the following components:

- .env file:** Contains configuration for a backend service, including Groq API settings and Jira integration details.

```
1 PORT=8091
2 CORS_ORIGIN=http://localhost:5173
3
4 # Groq Configuration
5 groq_API_BASE=https://api.groq.com/openai/v1
6 groq_API_KEY=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
7 groq_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
8
9 # Jira Configuration
10 JIRA_BASE_URL=https://suhasinworkspace-40182683.atlassian.net
11 JIRA_EMAIL=suhasiniganesh34@gmail.com
12 JIRA_API_TOKEN=ATATT3xFFGF0Y3AT32ocN5wd12AaCdPVLkWxFPy5fdgpH8Xobip
```
- Terminal:** Shows the command `npm run dev` being executed in the backend directory, with a response from the Jira API.

```
PS D:\Gen AI\Materials\user-story-to-tests (2)\user-story-to-tests > npm run dev
/rest/api/3/statuscategory/2",
  "id": 2,
  "key": "new",
  "colorName": "blue-gray",
  "name": "To Do"
}
```
- Chat Interface:** Displays a summary of the Jira integration implementation plan, listing created and updated files for both backend and frontend.

JIRA INTEGRATION IMPLEMENTATION PLAN
Perfect! ☒ Jira Integration Implementation Complete!

All files have been successfully created and configured. Here's what's been set up:

Backend Files Created:

- ☒ `jiraConfig.ts` - Jira configuration with environment variables
- ☒ `jiraClient.ts` - Jira API client service
- ☒ `jira.ts` - Jira API endpoints
- ☒ Updated `schemas.ts` - Added Jira data schemas
- ☒ Updated `server.ts` - Registered Jira routes

Frontend Files Created:

- ☒ `ProjectSelector.tsx` - Project dropdown component
- ☒ `StorySelector.tsx` - Story dropdown component
- ☒ Updated `types.ts` - Added Jira types
- ☒ Updated `api.ts` - Added Jira API functions
- ☒ Updated `App.tsx` - Integrated Jira components with auto-fill

Configuration Updated:

- ☒ `.env`

Describe what to build

package.jsonTS jira.ts.env frontend.env \.jira-vit

.env

1 PORT=8091
2 CORS_ORIGIN=http://localhost:5173
3
4 # Groq Configuration
5 groq_API_BASE=https://api.groq.com/openai/v1
6 groq_API_KEY=gsk_Gs91jL5kAmnXvOB3dcJtWGdyb3FYCWkLWwdY81czEX1N3S9iozA
7 groq_MODEL=meta-llama/llama-4-scout-17b-16e-instruct
8
9 # Jira Configuration
10 JIRA_BASE_URL=https://suhasinisworkspace-40182683.atlassian.net
11 JIRA_EMAIL=suhasiniganesh34@gmail.com
12 JIRA_API_TOKEN=ATATT3xFGF0Y3AT32ocN5wd12AaCdPVLkwxFPy5fdgph8XobipT3ow36

PROBLEMS1OUTPUTDEBUG CONSOLETERMINALPORTS

PS D:\Gen AI\Materials\user-story-to-tests (2)\user-story-to-tests > npm run dev
/rest/api/3/statuscategory/2",
"id": 2,

CHAT

JIRA INTEGRATION IMPLEMENTATION PLAN

- Backend: http://localhost:8091
- Frontend: http://localhost:5173

Features Now Available:

- Jira Project Dropdown - Auto-loads all projects from your Jira instance
- User Story Dropdown - Loads stories when project is selected
- Auto-Population - Story title and description auto-fill the form
- Test Generation - Generate test cases from Jira stories
- Error Handling - Clear error messages for configuration issues

Happy Testing! 🎉

Let me know when you're ready to test, or if you need any adjustments to the implementation!

Front End Accessed:

localhost:5173

★ Jira Integration

Jira Project

Scrum (SCRUM)

User Story

-- Select a User Story --

Story Title *

Enter the user story title...

Description

Additional description (optional)...

Acceptance Criteria *

Enter the acceptance criteria...

localhost:5173

Jira Integration

Jira Project

Scrum (SCRUM)

User Story

-- Select a User Story --

-- Select a User Story --

SCRUM-52: Employee Profile Employee Self

SCRUM-51: AHC Reports Reports AHC

SCRUM-50: Secretary Dashboard Doctor Calendar

SCRUM-49: Masters Others Approval Setup

SCRUM-48: Employee Profile Basic Details

SCRUM-47: Doctor Dashboard Op Dashboard

SCRUM-46: Reports Admin Reports Others

SCRUM-45: Consultant Dashboard My Appointments

SCRUM-44: Billing Execution Op Bill

SCRUM-43: Radiology Transactions Reporting Worklist

SCRUM-42: Reports Bb Report Home

SCRUM-41: Doctor Activities Doctor Note

SCRUM-40: Wards Dashboard View Request

SCRUM-39: Nurse Dashboard View Request

SCRUM-38: Daycare Dashboard View Request

SCRUM-37: Consultant Dashboard View Request

SCRUM-36: Radiology Transactions Front Office

SCRUM-35: Radiology Transactions Technicians Worklist

localhost:5173

Jira Integration

Jira Project

Scrum (SCRUM)

User Story

SCRUM-48: Employee Profile Basic Details

Story Title *

Employee Profile Basic Details

Description

As a user, I want to employee profile basic details so that the related functionality works as expected.

Acceptance Criteria *

Given the system is operational, when the user performs the intended action, then the system should respond accurately and meet functional expectations.

Generated Test Cases

20 test case(s) generated • Model: meta-llama/llama-4-scout-17b-16e-instruct • Tokens: 2565

Test Case ID	Title	Category	Priority	Expected Result
▶ TC-001	View Employee Profile Basic Details	Positive	High	Employee basic details are displayed accurately
▶ TC-002	Update Employee Profile Basic Details	Positive	High	Employee basic details are updated successfully
▶ TC-003	Search Employee Profile	Positive	Medium	Employee profile is displayed
▶ TC-004	Invalid Employee ID	Negative	Medium	Error message is displayed
▶ TC-005	Empty Employee Profile	Edge	Low	Employee profile is empty
▶ TC-006	Update Employee Profile with Invalid Data	Negative	Medium	Error message is displayed
▶ TC-007	View Non-Existent Employee Profile	Negative	Low	Error message is displayed
▶ TC-008	Update Employee Profile with Duplicate Data	Negative	Medium	Error message is displayed
▶ TC-009	Employee Profile with Special Characters	Edge	Low	Employee basic details are updated successfully
▶ TC-010	Employee Profile with Long Strings	Edge	Low	Employee basic details are updated successfully

Exported to excel



employee_profile_ba
sic_details_testcases_2